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*****
* MEADE #506 CCS Protocol *
*****
AdNovea® 2026
```

The bus handles a pseudo I2C protocol.

```
MASTER = Autostar
SLAVE   = #506 CCS
```

Clock (~1.5kHz) is ALWAYS generated by the MASTER.
MASTER ALWAYS send the header.

READ (Slave send message)

```
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[header][len][payload]
```

```
DIR
-> header      = 0x03 0x11 0x03
<- len        = len of payload
<- payload     = bytes of message
```

After 24 clock cycles, the SLAVE takes the DATA line control for sending the message length and optionnaly the message bytes.

If the length is greater than 0, the MASTER generated the clock cycles required for sampling the SLAVE message bytes.

if there is no payload, the len returns 0x00.
This can be viewed as polling message that occurs ever ~100ms

NOTE: When the DATA line stays open (pulled-up) after the header, the len is read as 0xFF.

The MASTER takes this as an error and a 5 seconds pause is generated by the MASTER.

WRITE (master send message)

```
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[header][payload][terminator]
```

```
DIR
-> header      = 0x05 0x11 0x01
-> payload     = 2-byte of message
-> terminator  = 0x00
```

Depending of the length of the MASTER message, the sequence above is repeated as many time as necessary.

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```

Autostar data are sampled on CLK Rising edge
CSS module data are sampled on CLK Falling edge

	Autostar	Autostar
Polling	Read	Write

#494 Autostar:

Add Source	0x03	0x03	0x05	-+
Add Destination	0x11	0x11	0x11	+ Header
Command/Action[1]	0x03	0x03	0x01	-+

#506 CCS:

Length/Status	0x00	LEN	BYTE1
Payload		PAYLOAD	BYTE2
Terminator		<none>	0x00

Addresses:

0x03	Handset Meade Autostar #494
0x05	#506 CCS Meade module
0x11	???

Commands:

0x03	Polling #506 that replying 0x00=none or return LEN + PAYLOAD
0x01	Autostar send 2-Byte + 0x00 (repeat frame until all chars are sent)

#506 CCS -> #494 Autostar

23 Stream: #:GD# -> Raw Packet (9 bytes): 03 11 03 05 23 3A 47 44

23 Stream: #:GR# -> Raw Packet (9 bytes): 03 11 03 05 23 3A 47 52

#494 Autostar -> #506 CCS

31 DF 00 05 11 01 31 31 00 05 11 01 3B 31 00 05 11 01 31 23 00 03 11
-> Raw Packet (32 bytes): 05 11 01 2B 31 00 05 11 01